Milestone 2

Project ID : CMPG325-2026-072

Client ID : CLI-072

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Module : CMPG 325 - Computer Networks

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Client Overview:

Organisation : Thabang Building Contractors (Klerksdorp)

Industry : Construction

Assigned Addressing Block:172.30.46.0/23

Submitted Items for Review:

This Milestone 2 submission includes the following evidence:

1. Working Packet Tracer File

- Location: 04_Packet_Tracer/CMPG325_Milestone2_MENGWAI.pkt

- Description: Fully configured network with VLSM, inter-VLAN routing, and all VLANs implemented.

2. Assigned Feature: VLSM Implemented

- All 7 VLANs configured with correct subnets and gateways

- SVIs on Core L3 Switch match the VLSM addressing plan exactly

- Inter-VLAN routing verified and working

3. Testing Evidence

Location: 05_Testing_Evidence/

- Screenshots of successful pings (same-VLAN, inter-VLAN, CR2)

- show ip route verification on Core Switch

- show vlan brief verification on access switches

4. Updated GitHub Portfolio

- New files added with meaningful commits

- Repository structure maintained and organised

Network Implementation Summary:

VLANs Configured

VLAN ID	Name	Subnet	Gateway	Status
10	MGMT	172.30.46.192/27	172.30.46.193	Active
20	DESIGN	172.30.46.160/27	172.30.46.161	Active
30	SALES	172.30.46.224/28	172.30.46.225	Active
40	SITE_OPS	172.30.46.128/27	172.30.46.129	Active
50	SERVERS	172.30.46.240/29	172.30.46.241	Active
60	GUEST	172.30.46.0/26	172.30.46.1	Active
70	FLOOR2	172.30.46.64/26	172.30.46.65	Active
99	NATIVE	(trunk only)		Active

Devices Configured

Device	Model	Role
Edge_Router	Cisco 2811	WAN connection, default route
Core_L3_SW	Cisco 3560	Inter-VLAN routing, SVIs, DHCP
GroundFloor_SW	Cisco 2960	Access switch for ground floor
FirstFloor_SW	Cisco 2960	Access switch for CR2 floor
Server1	Server	VLAN 50
Dept_PC1-4	PCs	VLANs 10, 20, 30, 40
PC-Floor2	PC	VLAN 70 (CR2)
AP1-Guest	Access Point	VLAN 60

Testing Summary

Test	Command Used	Result
Same-VLAN ping	`ping 172.30.46.193` from Dept_PC1	Pass
Inter-VLAN ping	`ping 172.30.46.162` from Dept_PC1	Pass
CR2 connectivity	`ping 172.30.46.66` from Dept_PC1	Pass
Routing table	`show ip route` on Core_L3_SW	All subnets present
VLAN verification	`show vlan brief` on GroundFloor_SW	Correct assignments

VLAN verification	`show vlan brief` on FirstFloor_SW	Correct assignments

VLSM Implementation Details

Address Block: 172.30.46.0/23

VLAN	Subnet	Mask	Usable Hosts	Gateway
60 (GUEST)	172.30.46.0	/26	62	172.30.46.1
70 (FLOOR2)	172.30.46.64	/26	62	172.30.46.65
40 (SITE_OPS)	172.30.46.128	/27	30	172.30.46.129
20 (DESIGN)	172.30.46.160	/27	30	172.30.46.161
10 (MGMT)	172.30.46.192	/27	30	172.30.46.193
30 (SALES)	172.30.46.224	/28	14	172.30.46.225
50 (SERVERS)	172.30.46.240	/29	6	172.30.46.241
WAN Link	172.30.46.248	/30	2	—
Router-Switch Link	172.30.46.252	/30	2	—

Why This Is Appropriate:

- Each subnet is sized close to its actual requirement (no group is given a mask larger than it needs), which is the core justification VLSM requires over fixed-length subnetting.

- The two largest groups (Guest Wi-Fi and the CR2 floor) were given the most headroom (/26, 62 usable each).

- The WAN link uses a /30, the smallest usable subnet, since a point-to-point link only ever needs two addresses.

- Significant free space remains in the block for future subnets.

Design Constraint (Management Always Online)

Requirement : Management requires internet access even when the staff network is restricted.

Implementation : The design uses VLAN segmentation with an extended ACL on the edge router's WAN-facing interface. The ACL permits the Management subnet (172.30.46.192/27) to any destination before any deny rule that applies to other staff VLANs.

Status : ACL will be implemented in the final phase.

Change Request CR2 (Additional Floor)

Requirement : The client takes over an additional floor/area and needs coverage there.

Implementation :

- VLAN 70 (FLOOR2) was created with subnet 172.30.46.64/26
- FirstFloor_SW connects via trunk to Core_L3_SW

- PC-Floor2 and AP1-Guest are connected and functional
- Existing ground-floor VLANs continue to function unchanged

    ❖ Note: The First-Floor switch is connected to FastEthernet0/1 on the Core L3 Switch because the 3560 model only provides two Gigabit ports.

Status : Fully implemented and tested

Notes

- DHCP will be added in the next phase (final submission)

- ACL for design constraint (Management always online) will be added next

- All configurations are saved and reproducible in the .pkt file